Answer key: Lesson 4, Odd + Odd: Always, Sometimes, Never

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the pairs. Fill in the missing numbers.

Answer: 6 counters: 0 left over; 7 counters: 1 left over

6 counters make 3 pairs with none left over. 7 counters make 3 pairs and 1 counter is left over.

Watch for: Writing the number of pairs instead of the number left over.; Not noticing the single counter at the end.

S2. Pair up counters or draw dots in pairs. Circle Odd or Even for each number.

Answer: 4 even; 9 odd; 10 even; 13 odd

4 makes 2 pairs: even. 9 makes 4 pairs and 1 left over: odd. 10 makes 5 pairs: even. 13 makes 6 pairs and 1 left over: odd.

Watch for: Calling a big number even because it is big.; Looking at the first digit instead of the last digit.

S3. Count in 2s. Colour each number you say. Then fill in the missing number.

Answer: The last digits are 2, 4, 6, 8 and 0

The numbers are 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. The last digits are 2, 4, 6, 8, 0, 2, 4, 6, 8, 0. The missing digit is 0.

Watch for: Colouring every number.; Skipping 10 or 20.

S4. Which number is even? Write its number in the bracket.

Answer: (3) 14

7 is odd (last digit 7). 11 is odd (last digit 1). 14 is even (last digit 4). 15 is odd (last digit 5). So the even number is 14.

Watch for: Choosing the biggest number.; Choosing 11 because it has two same digits.

S5. Look at the 11 counters in pairs. Fill in the blank. Then circle Odd or Even.

Answer: 1 counter has no partner; 11 is odd

11 counters make 5 pairs. 1 counter is left without a partner, so 11 is odd.

Watch for: Writing 5 (the number of pairs).; Writing 0.

S6. Read each statement. Decide if it is True or False.

Answer: True, False, True, False

6 is even: True. 9 is odd, not even: False. 15 is odd: True. 20 is even, not odd: False.

Watch for: Marking 9 is even as true.; Thinking 20 is odd because it is big.

Think harder (page 3)

T1. Mei says: an odd number plus an even number is always odd. Test it. Fill in the blanks.

Answer: Totals 11, 13, 9, all odd; Yes, the claim works for these three

6 + 5 = 11 (odd). 10 + 3 = 13 (odd). 8 + 1 = 9 (odd). All three totals are odd, so the claim works for these three sums. Three tests are good evidence, not a proof.

Watch for: Saying Yes means the claim is true for every number.; Adding wrongly (6 + 5 = 12).

T2. Circle Odd or Even for each number.

Answer: 57 odd; 80 even; 34 even; 91 odd

57 ends in 7: odd. 80 ends in 0: even. 34 ends in 4: even. 91 ends in 1: odd.

Watch for: Looking at the first digit (the tens).; Thinking 0 is odd.

T3. Ravi says: an odd number plus an odd number is always odd. Test it with 9 + 3. Fill in the blanks.

Answer: 9 + 3 = 12; No, the claim is not true; odd + odd is Never odd (pairs picture)

9 + 3 = 12, and 12 is even. So Ravi is not right. Each odd number has one counter left over, and the two left-over counters make a pair, so odd plus odd is never odd.

Watch for: Writing 11 or 13.; Saying Yes because 9 and 3 are both odd.

T4. Can you make each number by adding two odd numbers? Circle Yes or No.

Answer: 12 yes (5 + 7); 15 no; 20 yes (9 + 11)

12 = 5 + 7 (yes). 15 is odd, and two odd numbers always make an even total, so no. 20 = 9 + 11 (yes).

Watch for: Trying only one pair for 15 and giving up too soon.; Choosing Yes for every number.

Go further (page 4)

G1. Look at 3 + 5. Draw 3 counters and 5 counters in pairs. Fill in the missing numbers.

Answer: 1 left over, 1 left over, 1 pair

3 counters make 1 pair and 1 left over. 5 counters make 2 pairs and 1 left over. The 2 left-over counters make 1 new pair, so 3 + 5 has none left over.

Watch for: Forgetting that the two left-over counters pair up.; Drawing the wrong number of counters.

G2. Add three odd numbers. Circle Odd or Even for each total.

Answer: 3 + 5 + 7 = 15 odd; 1 + 1 + 1 = 3 odd; 9 + 3 + 5 = 17 odd; odd + odd + odd is Odd

3 + 5 = 8 (even), 8 + 7 = 15 (odd). 1 + 1 = 2, 2 + 1 = 3. 9 + 3 = 12, 12 + 5 = 17. Odd + odd is even, and even + odd is odd, so the total is odd.

Watch for: Saying even because odd + odd is even.; Adding wrongly.

G3. Look at 4 + 6. Draw 4 counters and 6 counters in pairs. Fill in the blank, then finish the sentence.

Answer: 0 counters are left over; even + even is even because no counter is left over

4 makes 2 pairs and 6 makes 3 pairs. Neither has a counter left over, so together they make 5 pairs and none is left over. The total, 10, is even.

Watch for: Counting 4 + 6 = 10 without looking at the pairs.; Writing odd.

Investigation (page 5): Numbers next to each other

Always odd: 7, 11, 17, 21. One of the two numbers is odd and one is even, and odd + even is odd. Another way: a number plus the next number is a double plus 1, and a double has no counter left over, so one more is odd.

Look for: A child who tries several pairs, records odd or even, and says always. Credit any clear reason: one number is odd and one is even, or it is a double and 1 more (from last lesson, if the child was there). Do not mark a child down for stopping at a few tests.

Exit ticket (page 5)

E1. Is 17 odd or even? Circle Odd or Even.

Answer: Odd

17 makes 8 pairs and 1 counter is left over. The last digit is 7: odd.

Watch for: Choosing Even because 17 is a big number.; Looking at the tens digit.

E2. Which number is even? Write its number in the bracket.

Answer: (3) 18

9, 13 and 21 end in odd digits. 18 ends in 8: even.

Watch for: Choosing 21.; Choosing the first option.

E3. Can you make 14 by adding two odd numbers? Circle Yes or No.

Answer: Yes

7 + 7 = 14 (also 5 + 9). 14 is even, and odd plus odd is even.

Watch for: Saying No because 14 is even.; Not trying 7 + 7.

Sneak-peek puzzles (not in the workbook)

1. 1
2. No. 5 + 7 = 12 counters make 6 pairs. The one left over from the 5 and the one left over from the 7 make a pair.